PROJECT EVALUATION SHEET

Project Prototype Part 1 | Midterm

B.Sc. in Computer Science and Engineering (CSE) Program

Course code	CSC 3215	**Course title**	Web Technologies
Semester	Fall 2026-27	**Section**	J
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Course Outcome

CO3 [PO-c2]	**Develop a multi-tier web application for a targeted society**
AI Policy	***The use of AI tools is strongly discouraged. A student who still needs an AI tool must get the course teacher's consent in advance and must state which tool was used and the exact parts of the work it was used for. Every student must be able to explain all of the submitted work; any content generated by AI that the student cannot explain will be marked as Missing/Incorrect (0).***

Evaluation (Course Teacher's Use Only)

Assessment Attribute/Criteria	Component	Marks (0–5)	Criterion Mark
Front-End Design and Code Structure (10 marks)	Pages & Forms (HTML)		
	UI Design (CSS)		
	Code Structure (HTML/CSS/JS separation)		
Feature Implementation and Explanation (10 marks)	JS Form Validation		
	DOM Manipulation		
	VIVA (Code Explanation)		
		CO3 Total	

Project Details

Project Title	**MetroSheba — Online Metro Ticketing Platform for Dhaka MRT Line-6**

Project Description

Describe the targeted society or users, the problem the project solves and the main pages, forms and features of the prototype.

Targeted society and users. Daily commuters of the Dhaka Metro Rail (MRT Line-6, Uttara North to Motijheel) and the staff who administer ticket sales. Passengers currently queue at station counters and vending machines during peak hours, which wastes time and causes crowding. MetroSheba lets a commuter buy a ticket online before reaching the station, and gives the authority a live view of sales.

Problem the project solves. There is no single place where a commuter can register, buy a metro ticket, see the fare before paying and keep a record of past journeys. Counter staff also have no quick way to see which routes sell most or how much revenue a day produced. The application solves both sides with one multi-tier system.

Architecture. A three-tier MVC structure. Views hold only HTML; styling lives in four external stylesheets; behaviour lives in ten external JavaScript files; controllers validate input and enforce the business rules; models hold every MySQL query using prepared statements. No page writes SQL and no model prints HTML.

Main pages and forms. Home; Registration form with OTP verification; Login form; Passenger Dashboard with the ticket purchase form; My Tickets purchase history; Profile view with picture upload; Edit Profile form; Admin Dashboard for customer management; and the Admin Ticket Sales report.

JS form validation. Every form is validated in the browser before submission — name (two words, no digits), email format, 11-digit mobile starting with 01, numeric NID, date of birth, password length and a 6-digit OTP. Each rule is repeated in PHP, because the client-side copy can be bypassed.

DOM manipulation. The station dropdowns and route map are generated at runtime; the fare updates live as the stations or quantity change and the chosen leg is highlighted; the ticket receipt, the purchase-history table, the customer table and the busiest-routes bar chart are all built with createElement and textContent.

AJAX and MySQL. Ticket purchase, OTP delivery, ticket cancellation, picture upload and all admin edit/delete actions run through XMLHttpRequest against PHP endpoints returning JSON, so no action reloads the page. Data is stored in MySQL (users and metroshebatickets), created by database/metrosheba.sql.

Fare rule. The fare rises by 10 BDT per station with a 20 BDT minimum, so Uttara North to Motijheel (14 stations) costs 140 BDT per ticket. The rule is defined once on the server and sent to the browser, and the server recalculates the total on purchase, so a tampered form cannot buy a cheaper ticket.

Snap of the UI

Attach screenshots of the main pages and forms of the prototype (for example, the home page and a form with JavaScript validation).

Registration Form — JavaScript Validation

Each field is checked in the browser and the failing input is marked before the form can be submitted.



